

Technology Company Performance Analysis with Automation

Project Overview:

- Collected GitHub repository data from leading technology companies using the GitHub REST API.
- Used Python to collect, clean, preprocess, and transform the repository data.
- Stored the cleaned data in a MySQL database for structured data management and SQL analysis.
- Used SQL to query, validate, and analyze the repository data.
- Created a Google Sheets dataset to maintain the processed repository data.
- Automated the data collection and update process using n8n workflow automation.
- Configured n8n to collect repository data, process it, update Google Sheets, and send email notifications automatically.
- Connected the processed data to Power BI for dashboard creation and business reporting.
- Performed Exploratory Data Analysis (EDA) using Pandas, NumPy, Matplotlib, and Seaborn.
- Analyzed repository metrics such as Stars, Forks, Watchers, Open Issues, Programming Language, Repository Age, and Repository Status.
- Created statistical analysis and visualizations to identify repository performance, developer engagement and technology trends.
- Built an end-to-end automated data analytics workflow from data collection to dashboard reporting.
- Generated meaningful business insights to support data-driven decision-making.

Data Source:

Attribute	Details
Dataset Name	Technology Company Performance Analysis Dataset
Primary Source	GitHub REST API (Public Repository Data)
Data Collection	Collected automatically using Python and the GitHub REST API
Automation Tool	n8n Workflow Automation
Intermediate Storage	Google Sheets
Database	MySQL

Dashboard Tool	Power BI
Dataset Format	CSV
Number of Records	1,586 Repositories
Number of Attributes	13
Companies Covered	Microsoft, Google, Oracle, IBM, SAP, Salesforce, Snowflake and SAS
Domain	Information Technology / Open-Source Software Analytics
Timeline	Repository data collected from 2021–2026

Problem Statement:

- Leading technology companies maintain thousands of open-source repositories on GitHub.
- Collecting repository data manually is time-consuming and repetitive.
- Repository data must be cleaned, stored, and updated before analysis.
- Manual data updates may lead to delays and data inconsistency.
- Organizations need an automated workflow to collect and manage repository data efficiently.
- Business users require interactive dashboards to monitor repository performance and developer engagement.
- This project automates data collection, processing, and reporting using GitHub REST API, Python, SQL, Google Sheets, n8n and Power BI to generate meaningful business insights.

Objectives:

- To collect GitHub repository data from leading technology companies using the GitHub REST API.
- To clean, preprocess, and transform the collected data using Python.
- To store the processed data in a MySQL database for structured analysis.
- To automate the data collection and update process using n8n.
- To maintain the latest repository data in Google Sheets.
- To analyze repository performance using Exploratory Data Analysis (EDA) and statistical analysis.
- To create an interactive Power BI dashboard for business reporting.
- To identify repository popularity, developer engagement, and programming language trends.
- To generate meaningful business insights for data-driven decision-making.

 To develop an end-to-end automated data analytics workflow.

Attribute Details:

Attribute Name	Description
Company	Name of the technology company that owns the GitHub repository.
Repository Name	Name of the GitHub repository.
Programming Language	Primary programming language used in the repository.
Stars	Number of GitHub stars received, representing repository popularity.
Forks	Number of times the repository has been forked by developers.
Watchers	Number of users watching the repository for updates.
Open Issues	Number of unresolved issues currently available in the repository.
Repository Size (KB)	Storage size of the repository in kilobytes.
License	Software license associated with the repository.
Archived	Indicates whether the repository is archived or actively maintained.
Created Date	Date on which the repository was originally created.
Updated Date	Most recent date when repository activity was recorded.
Repository URL	Direct GitHub URL of the repository.

Tools & Technologies:

Category	Technologies / Tools
Programming Language	Python
Development Environment	Google Colab
Data Collection	GitHub REST API
API Library	Requests
Data Processing	Pandas, NumPy
Database	MySQL
Automation	n8n
Cloud Storage	Google Sheets

Dashboard	Power BI
Dataset Format	CSV

Project Workflow :

- 🌐 Collect repository data using the GitHub REST API.
- 🌐 Process and clean the data using Python.
- 🌐 Store the cleaned data in MySQL.
- 🌐 Automate the workflow using n8n.
- 🌐 Update the processed data in Google Sheets.
- 🌐 Create reports and dashboards in Power BI.
- 🌐 Generate business insights from the analyzed data.

Data Collection :

- 🌐 Repository data was collected from the GitHub REST API.
- 🌐 Python was used to send API requests and retrieve repository information.
- 🌐 Personal Access Token (PAT) authentication was used to access the GitHub API.
- 🌐 Repository data was collected from Microsoft, Google, Oracle, IBM, SAP, Salesforce, Snowflake and SAS.
- 🌐 The collected data included repository name, programming language, stars, forks, watchers, open issues, repository size, license, repository status, created date, updated date and repository URL.
- 🌐 The collected data was processed through the n8n automation workflow.
- 🌐 The processed data was automatically updated in Google Sheets.
- 🌐 The final dataset was stored in MySQL and used for Power BI dashboard reporting.

Data Cleaning :

- 🌐 Verified the dataset structure using Python.
- 🌐 Examined the first and last records of the dataset.
- 🌐 Checked the total number of rows and columns.
- 🌐 Identified missing values in the dataset.
- 🌐 Replaced missing values in the Programming Language column with "Unknown".
- 🌐 Replaced missing values in the License column with "No License".

- 🌐 Converted Created Date and Updated Date into DateTime format.
- 🌐 Removed duplicate repositories based on company and repository name.
- 🌐 Standardized repository status as Active and Archived.
- 🌐 Converted numerical fields into the correct data types.
- 🌐 Verified the cleaned dataset before storing it in MySQL.

MySQL Database :

- 🌐 Created a MySQL database to store the cleaned repository data.
- 🌐 Imported the processed CSV dataset into MySQL.
- 🌐 Created a table with appropriate data types for all repository attributes.
- 🌐 Verified the imported records after loading the dataset.
- 🌐 Used SQL queries to retrieve and analyze repository data.
- 🌐 Applied filtering, sorting, grouping, and aggregation queries.
- 🌐 Performed joins, views, stored procedures, triggers, and CASE statements where required.
- 🌐 Validated the dataset before using it for automation and dashboard reporting.
- 🌐 Used MySQL as the central data source for the automation workflow.

n8n Workflow Automation:

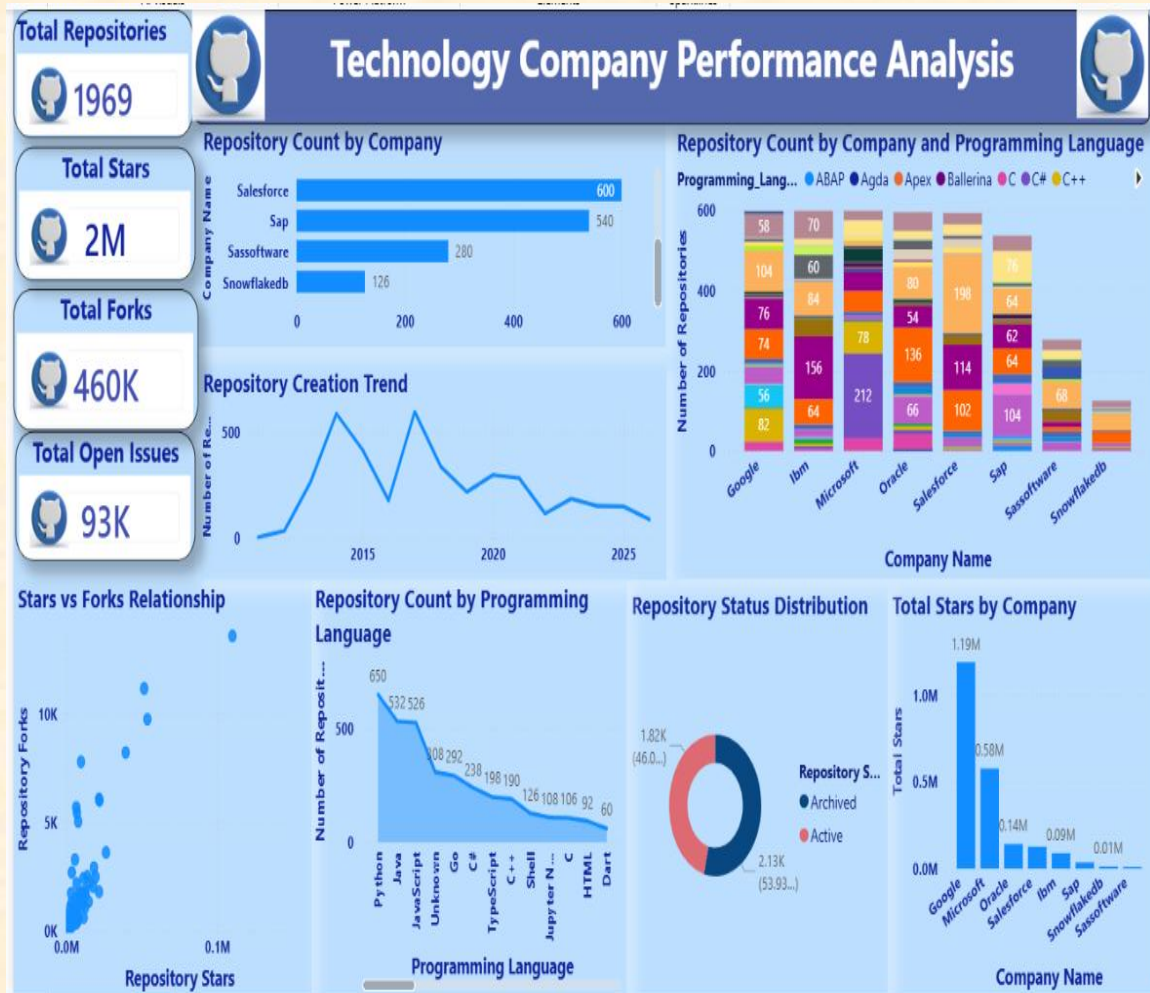
- 🌐 Created an automated workflow using n8n.
- 🌐 Used a Schedule Trigger to run the workflow automatically.
- 🌐 Retrieved GitHub repository data using the GitHub REST API.
- 🌐 Processed and transformed the repository data using JavaScript Code nodes.
- 🌐 Removed duplicate repository records before updating the dataset.
- 🌐 Cleaned and validated the repository data within the workflow.
- 🌐 Automatically appended the processed data to Google Sheets.
- 🌐 Prepared the updated dataset for Power BI reporting.
- 🌐 Reduced manual effort by automating the data collection and update process.

Google Sheets Integration :

- 🌐 Connected Google Sheets with the n8n workflow.
- 🌐 Stored the processed GitHub repository data in Google Sheets.
- 🌐 Automatically appended new repository records to the spreadsheet.
- 🌐 Maintained a structured dataset for reporting and analysis.
- 🌐 Used Google Sheets as an intermediate data storage platform.

- Verified the updated records before dashboard reporting.
- Enabled easy access to the latest repository data.

PowerBI Dashboard:



- Connected Power BI with the processed repository dataset.
- Imported the latest repository data for reporting.
- Created an interactive dashboard to visualize repository performance.
- Displayed key metrics such as Stars, Forks, Watchers, and Open Issues.
- Compared repository performance across technology companies.
- Analyzed programming language trends and repository status.
- Used filters and slicers for interactive data exploration.
- Generated business insights to support data-driven decision-making.

Business Insights :

- 🌐 The automated workflow successfully collected and processed GitHub repository data.
- 🌐 Google and Microsoft repositories showed higher developer engagement than the other companies.
- 🌐 JavaScript, Python, and Java were the most widely used programming languages.
- 🌐 Repository performance was evaluated using Stars, Forks, Watchers and Open Issues.
- 🌐 The interactive Power BI dashboard simplified repository performance monitoring.
- 🌐 Automation reduced manual effort and improved data consistency.
- 🌐 The project provided meaningful business insights to support data-driven decision-making.

Future Enhancements :

- 🌐 Integrate real-time GitHub API data refresh.
- 🌐 Automate scheduled data updates using n8n.
- 🌐 Expand the analysis by including more technology companies.
- 🌐 Develop predictive models for repository performance.
- 🌐 Build advanced interactive dashboards with additional KPIs.
- 🌐 Deploy the project as a web-based analytics platform.
- 🌐 Improve reporting with AI-powered business insights.

Conclusion :

- 🌐 This project successfully automated the GitHub repository data collection and reporting process.
- 🌐 Repository data was collected using the GitHub REST API and processed using Python.
- 🌐 The cleaned data was stored in MySQL and automatically updated through the n8n workflow.
- 🌐 Google Sheets was used as an intermediate data storage platform for reporting.
- 🌐 Power BI was used to create an interactive dashboard for repository performance analysis.
- 🌐 The automated workflow reduced manual effort and improved data accuracy and consistency.
- 🌐 The project demonstrates an end-to-end data engineering and business intelligence workflow.
- 🌐 Overall, the project provides a scalable solution for automated data collection, processing, and business reporting.